FAST-MVH: 28-Instance Canonical Benchmark & Leave-One-Out Study

Complete Unconstrained Evaluation of Dimensionality Reduction Breakdown in Multi-Valued Heuristic Search (M=4-8)

100% Bit-Identical Pareto Fronts

Zero Timeouts | Maya Wohlf Reference Oracle

Hardware: Intel Core i7-13620H | -O3 -std=c++20

ELIMINATION OF THE O(N²) BOTTLENECK

OVERALL SPEEDUP: 22.58X GMEAN

When Pareto frontiers scale into dense antichains, unindexed linear scans in Maya Wohlf's reference baseline collapse under quadratic scaling: on **grid_8x8_M8_rho0.0** (|Π*|=369,351), the baseline requires **89,469.1s (~24.85 hours)**. FAST-MVH solves the identical problem in **93.16s** (—**960.4x speedup**). Across all 28 canonical settings, total wall-clock time is slashed from **41.6 hours** to **12.1 minutes** while maintaining 100% cryptographic SHA-256 Pareto equivalence.

LEAVE-ONE-OUT ARCHITECTURAL HIERARCHY

Architectural Pillar	Rank	GMean Slowdown	Peak Slowdown
Goal Frontier K-d Tree	1	**3.38x**	**24.34x** (grid_8x8_M8)
Local Frontiers K-d Tree	2	**1.88x**	**7.77x** (grid_8x8_M8)
Local-First Early Pruning	3	**1.55x**	**3.11x** (grid10_M5_rho-0.4)
Fallback Residue Forest	4	**1.10x**	**2.27x** (grid10_M5_rho-0.4)

TABLE 1: COMPLETE 28-INSTANCE EVALUATION MATRIX ACROSS ALL CONFIGURATIONS (NATURAL COMPLETION - ZERO TIMEOUTS)

EVALUATED WITH ≥2 CLEAN SAMPLES PER CELL | SUB-TENTH-SECOND RUNS MARKED WITH †

Instance Name	Dom	M	ρ	Sols (|Π*|)	Expansions	Baseline	FAST-MVH	w/o Goal	w/o Local	w/o L-First	w/o Forest	Speedup
Panda_S242_G93	Robo	8	0.0	13	130	**0.01s**	**0.004s**	0.0s	0.0s	0.0s	0.0s	1.9x
Panda_S29_G93	Robo	8	0.0	50	280	**0.03s**	**0.010s**	0.0s	0.0s	0.0s	0.0s	3.5x
Panda_S29_G405	Robo	8	0.0	46	6,711	**0.60s**	**0.22s**	0.2s	0.2s	0.2s	0.1s	2.7x
Panda_Q0024	Robo	8	0.0	109	40,057	**7.91s**	**2.19s**	2.3s	2.3s	2.7s	2.3s	3.6x
Panda_Q0010	Robo	8	0.0	2,203	93,945	**244s**	**68.9s**	69.9s	77.9s	80.3s	67.8s	3.5x
Panda_Q0002_Roi	Robo	8	0.0	6,581	738,325	**2.5h**	**34.6s**	98.9s	48.0s	41.3s	32.8s	263.6x
bay_8d_2000 (s500)	Road	5	0.0	5,529	36,915	**8.59s**	**0.42s**	9.7s	0.5s	0.7s	0.4s	20.7x
bay_8d_2000 (s2000)	Road	5	0.0	28,200	192,161	**183s**	**1.71s**	14.8s	3.4s	2.4s	1.7s	107.2x
grid10_M4_rho0.0	Grid	4	0.0	1,889	47,756	**1.56s**	**0.54s**	0.7s	0.6s	0.9s	0.5s	2.9x
grid10_M4_rho-0.2	Grid	4	-0.2	5,080	93,274	**5.85s**	**1.33s**	1.9s	1.6s	1.9s	1.4s	4.4x
grid10_M4_rho-0.4	Grid	4	-0.4	9,474	147,304	**12.3s**	**2.28s**	3.4s	2.9s	3.4s	2.6s	5.4x
grid12_M4_rho0.0	Grid	4	0.0	16,147	283,658	**47.1s**	**4.95s**	8.6s	7.4s	8.3s	8.3s	9.5x
grid15_M4_rho0.0	Grid	4	0.0	34,459	1,117,374	**1268s**	**18.1s**	21.2s	21.8s	16.6s	17.3s	70.1x
grid10_M5_rho0.0	Grid	5	0.0	23,150	269,107	**145s**	**6.96s**	23.5s	12.9s	10.5s	8.3s	20.8x
grid10_M5_rho-0.2	Grid	5	-0.2	48,303	475,255	**819s**	**32.4s**	48.1s	69.7s	53.6s	40.5s	25.3x
grid10_M5_rho-0.4	Grid	5	-0.4	58,746	546,830	**2020s**	**5.03s**	16.3s	13.5s	15.7s	11.4s	401.2x
grid12_M5_rho0.0	Grid	5	0.0	98,070	1,425,808	**1.7h**	**137.7s**	489.3s	369.6s	192.4s	291.9s	43.9x
grid_6x6_M6_rho0.0	Grid	6	0.0	3,269	16,487	**5.07s**	**0.64s**	1.4s	0.9s	2.0s	0.7s	7.9x
grid_8x8_M6_rho0.0	Grid	6	0.0	71,405	295,885	**1.0h**	**28.0s**	256.9s	110.5s	47.9s	33.7s	129.5x
grid_5x5_M7_rho0.0	Grid	7	0.0	1,694	5,370	**0.94s**	**0.16s**	0.3s	0.2s	0.3s	0.2s	5.7x
grid_6x6_M7_rho0.0	Grid	7	0.0	8,987	38,515	**20.0s**	**2.67s**	11.3s	5.5s	6.5s	2.6s	7.5x
grid_7x7_M7_rho0.0	Grid	7	0.0	59,157	221,389	**2934s**	**23.1s**	187.9s	83.8s	43.3s	32.4s	126.8x
grid_8x8_M7_rho0.0	Grid	7	0.0	192,162	823,682	**7.7h**	**191.9s**	3122.3s	1340.5s	277.2s	217.2s	144.2x
grid_5x5_M8_rho0.0	Grid	8	0.0	5,877	13,819	**12.8s**	**0.78s**	3.0s	1.1s	1.3s	0.8s	16.4x
grid_6x6_M8_rho0.0	Grid	8	0.0	22,726	65,839	**221s**	**6.53s**	42.0s	13.9s	10.3s	5.7s	33.9x
grid_6x6_M8_rho-0.2	Grid	8	-0.2	28,284	83,866	**322s**	**10.9s**	43.4s	16.5s	11.8s	8.6s	29.6x
grid_7x7_M8_rho0.0	Grid	8	0.0	75,169	287,802	**1.5h**	**49.6s**	500.3s	200.5s	68.9s	49.7s	110.8x
grid_8x8_M8_rho0.0	Grid	8	0.0	369,351	1,521,453	**24.9h**	**93.2s**	2267.8s	724.0s	155.4s	95.5s	960.4x
OVERALL SUMMARY (28 Instances, M=4-8)				1,176,108	7,542,883	149,709.5s	724.8s	3.38x GM	1.88x GM	1.55x GM	1.10x GM	22.58x GM

Verification Note: Every completed cell verified bit-identical via SHA-256 against Maya Wohlf reference outputs. Zero artificial timeouts.
Speedup Metric: Arithmetic means used for instance timings; Geometric Means (GMean) used for aggregated ratios. Sub-tenth-second runs filtered from ratio GMeans.